Mastara SRS: Update Sections (changes after v3.2)

Date: 27 October 2026

This document holds the new and replaced requirements agreed after SRS v3.2 (25 October 2026). Each section says which part of the SRS it replaces or adds to. Items marked **To confirm** are proposals that still need a decision.

1. Purpose and Scope

This update covers four areas of the Mastara mobile app, API and Admin Panel:

1. The offline-first architecture.
2. Student registration and the student ID (MS number).
3. Purchase eligibility for bundles and single courses.
4. Content access, downloads and the paywall flow.

2. Offline-First Architecture

Replaces: the earlier download-first rule (every course must be downloaded before it can be opened).

- **OFF-01** The mobile app shall be offline-first: the device is the primary place the app reads from and writes to, and the network is used to sync when available.
- **OFF-02** Content files shall be stored in cloud storage, not inside the app package, so content can be updated without releasing a new app version.
- **OFF-03** Downloading shall be optional for online students and required only for offline use (see section 5).
- **OFF-04** Pending changes shall sync automatically in the background whenever connectivity returns.
- **OFF-05** For conflicting edits to the same record, the latest change shall win, unless a requirement in this document states otherwise.

3. Student Registration and Student ID

Replaces: the rule that the student ID is assigned automatically on registration, and any rule that registration writes to the central database.

3.1 Before payment (local only)

- **REG-01** A student shall be able to register, log in and edit their profile fully offline.
- **REG-02** The registration shall be stored only on the device. Nothing shall be written to the central database.
- **REG-03** The local account shall use a temporary local ID (a random UUID), not an MS number.
- **REG-04** A student who has not paid shall not exist on the server and shall not appear in the Admin Panel.
- **REG-05** The registration screen shall tell the student that the account is saved on the phone until they buy a bundle.
- **REG-06** The "Browse Free Content (No Registration)" button shall be removed.

3.2 At payment (server registration)

- **REG-07** The server shall create the student record only after a bundle payment is confirmed.
- **REG-08** The server shall assign the MS number (MS0001, MS0002 and so on, auto-incrementing) at that moment, using an atomic database sequence or transaction so two students can never receive the same number.
- **REG-09** The MS number shall be assigned by the server only. The app shall never generate it.
- **REG-10** The payment request shall include the local ID. The server shall treat the request as idempotent: a retry with the same local ID shall return the same MS number and shall not create a second student.
- **REG-11** If the phone number or email already belongs to a paid student, the server shall link the payment to that existing student and shall not create a new MS number.
- **REG-12** After the MS number is returned, the app shall replace the local ID with the MS number and start normal sync.
- **REG-13** Device binding shall start when the server student record is created.

- **REG-14** If payment is approved manually by Operations Admin, the MS number shall be created at approval, not when the receipt is submitted. Until then the student shall stay local with a "Payment pending" status. **To confirm:** whether payment confirmation is automatic (gateway) or manual (receipt approval).

3.3 Recovery

- **REG-15** A student who has not paid and loses the phone or reinstalls the app shall lose the local account. This is accepted behaviour.
- **REG-16** A paid student shall be able to log in on the server after reinstalling, recover their purchases and re-download their courses.

4. Purchase Eligibility

Adds to: the pricing and bundles section.

- **PUR-01** A student's first purchase shall be a bundle: a batch bundle, a semester bundle, or the full course.
- **PUR-02** A student who owns an active bundle may buy extra single courses.
- **PUR-03** A student who has not bought a bundle shall not be able to buy a single course.
- **PUR-04** A single-course payment shall never create a new student. Only a confirmed bundle payment creates the server student (REG-07).
- **PUR-05** The API shall enforce these rules itself. The server shall reject:
  - a single-course payment from a student without an active bundle, including an unregistered local student;
  - a purchase of a course the student already owns through a bundle.
- **PUR-06** In the app, single courses shall show a locked button with the message "Buy a semester or full bundle first to unlock single courses" when the student has no bundle. They shall not be hidden.
- **PUR-07** A course already included in the student's bundle shall be shown as "Included" and shall not offer a Buy button.
- **PUR-08** Super Admin shall be able to set bundle prices and single-course prices, and turn the single-course option on or off, in payment settings.
- **PUR-09 To confirm:** if a bundle is refunded or expires, single courses the student bought separately stay available, but the student cannot buy more single courses until they own an active bundle again.

- **PUR-10 To confirm:** which bundles unlock single-course purchases. The current assumption is that batch, semester and full course bundles all do.

5. Content Access and Downloads

Replaces: the earlier must-download rule and any earlier content access rule.

- **ACC-01** All content listings (courses, modules, notes, Amharic notes, exams, tests, COC) shall be visible to every student instantly, including offline.
- **ACC-02** An unpaid student shall be able to open only free content.
- **ACC-03** Paid content shall open only for students who have paid for it.
- **ACC-04** An online student shall be able to open content without downloading it.
- **ACC-05** An offline student shall be able to open only content that was downloaded earlier.
- **ACC-06** Each course item shall show Download or Delete. Delete clears the local cache and the item shows Download again.

5.1 Access summary

Student	Online	Offline
No purchase (local only)	Free content opens	Free content opens only if downloaded
Bundle or course purchased	Opens without downloading	Opens only if downloaded
Any student, content not paid for	Locked Content message	Locked Content message

5.2 Download requirements

- **DWN-01** Downloads shall resume after an interruption.
- **DWN-02** The app shall show progress and file size, and warn before large downloads on mobile data.
- **DWN-03** Students shall be able to download per course, and per chapter where supported.
- **DWN-04** Downloaded paid content shall be encrypted and tied to the bound device.

- **DWN-05** The entitlement ("this student has paid") shall be stored locally with an expiry and renewed when the student is online.
- **DWN-06** Each course shall have a content version number. When the server version is newer than the downloaded one, the app shall show "Update available".
- **DWN-07** Students shall be able to delete downloads to free storage.

6. Paywall Flow

Replaces: the Locked Content flow for restricted content.

- **PAY-01** Tapping restricted content (Modules, Notes, Amharic Notes, Exams, Tests, COC) shall show "Locked Content / You cannot access this content" with a Buy / Purchase Course button.
- **PAY-02** The Buy button shall open the Store at the correct screen: bundles for a student with no bundle, bundles or single courses for a student with an active bundle.
- **PAY-03** The permission check at every gate (chapter, exam, test, COC) shall be: free content, or the student has paid for it. The result applies in the same way online and offline, using the locally stored entitlement.

7. Sync Rules

- **SYN-01** Registration data is local until the first confirmed bundle payment (section 3).
- **SYN-02** After registration on the server, profile edits, progress and purchases shall sync automatically when connectivity returns.
- **SYN-03** Admin Panel, Mobile App and API shall show the same data for courses, contents, universities, programs, semesters, exams, tests and COCs.

8. Admin Panel Impact

- **ADM-01** The Students page, student counts and Student IDs shall include only paid students.
- **ADM-02** Student IDs shall start at MS0001 and increase in order as students are created at payment.
- **ADM-03** Payment settings shall include bundle prices, single-course prices and the single-course on/off option (PUR-08).

- **ADM-04** Operations Admin shall be able to approve receipts where payment is manual (REG-14).

9. Open Items to Confirm

1. Is payment confirmed automatically by a payment gateway, or approved manually from receipts? (REG-14)
2. Do batch, semester and full course bundles all unlock single-course purchases? (PUR-10)
3. What happens to single courses when a bundle is refunded or expires? (PUR-09)
4. Does "extra course" mean a course outside the student's bundle, such as one from another semester or program? This decides how the "Included" check works (PUR-07).